

Course Name: Computer Networks and Internet Protocol

Assignment - Week 9 (Jan 2026)

TYPE OF QUESTION: MCQ/MSQ

Number of questions: 10

Total mark: 10 X 1 = 10

QUESTION 1

Which component of SDN is responsible for communicating with the controller?

- a) Data plane
- b) Control plane
- c) Management plane
- d) Forwarding plane

Correct Answer: (b)

Detailed Solution: The control plane is responsible for communicating with the controller and managing the flow of network traffic

QUESTION 2

A subset of a network that includes all the routers but contains no loops is called _____

- a) Spanning tree
- b) Spider structure
- c) Spider tree
- d) Special tree

Correct Answer: (a)

Detailed Solution: A spanning tree is a subset of a network that connects all nodes while eliminating loops. In Ethernet networks, Spanning Tree Protocol (STP) creates a loop free logical topology among switches/bridges.



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QUESTION 3

Which classes are used for connection-less socket programming?

- a) Datagram Socket
- b) Datagram Packet
- c) Both Datagram Socket & Datagram Packet
- d) Server Socket

Correct Answer: (c)

Detailed Solution: Datagram is basically some information travelling between the sender and the receiver, but there is no guarantee of its content, arrival or arrival time. Datagram Socket, Datagram Packet are used for connection-less socket programming, while Server Socket is used for connection-oriented socket programming.

QUESTION 4

Which of the following is NOT a function of a router?

- a) Path determination
- b) Packet forwarding
- c) Packet filtering
- d) Packet retransmission

Correct Answer: (d)

Detailed Solution: Packet retransmission is a function of the transport layer, specifically the TCP protocol. Routers do not perform packet retransmission, but rather forward packets to their destination based on the routing table and other factors.

QUESTION 5

What is the role of the OpenFlow protocol in SDN?

- a) To enable communication between the control plane and data plane
- b) To provide encryption for network traffic
- c) To monitor network performance and traffic
- d) To manage network security

Correct Answer: (a)

Detailed Solution: The OpenFlow protocol is used to enable communication between the control plane and data plane in an SDN environment.

QUESTION 6



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Which type of Ethernet framing is used for TCP/IP and DEC net?

- a) Ethernet 802.3
- b) Ethernet 802.2
- c) Ethernet II
- d) Ethernet SNAP

Correct Answer : (c)

Detailed Solution: The Ethernet 802.3 framing is used for NetWare versions 2 to 3.11, and the Ethernet 802.2 framing is used for NetWare versions 3.12 and later plus OSI routing, Ethernet II is used with TCP/IP and DEC net, and Ethernet SNAP is used with TCP/IP and AppleTalk. The type field in Ethernet 802.2 frame is replaced by a length field in Ethernet 802.3.

QUESTION 7

Which of the following is not true regarding Forward Information Base(FIB)?

- a) The FIB mirrors the Routing Information Base(RIB)
- b) Responsible for merging the data plane and control plane together
- c) Routing decisions are taken from FIB and not RIB
- d) It is stored in TCAM Hardware

Correct Answer: (b)

Detailed Solution: The presence of FIB allows the Data plane and Control Plane to be separated thus allowing them to run independently.

QUESTION 8

What is the role of the northbound interface in SDN?

- a) To manage and configure network devices
- b) To enable communication between the controller and applications
- c) To monitor network performance and traffic
- d) To provide network services to end users

Correct Answer: (b)

Detailed Solution: The northbound interface is used to enable communication between the controller and applications in an SDN environment. It provides a standardized interface that allows applications to request network services and receive information about the network state.

QUESTION 9



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A network 192.168.10.0/24 needs at least 12 subnets. Which subnet mask should be used?

- a) 255.255.255.224
- b) 255.255.255.240
- c) 255.255.255.192
- d) 255.255.255.248

Correct Answer: (b)

Detailed Solution: $2^4=16$ subnets - borrow 4 bits, new prefix = /28, i.e, 255.255.255.240

QUESTION 10

Which of the following statements are true regarding the command : *ip route 172.16.4.0 255.255.255.0 192.168.4.2* ?

- I. The command is used to establish a static route.
- II. The default administrative distance is used.
- III. The command is used to configure the default route.
- IV. The subnet mask for the source address is 255.255.255.0.

- a) I and II
- b) II and IV
- c) III and IV
- d) I, II, III and IV

Correct Answer: (a)

Detailed Solution: Although answer d almost seems right, it is not; the mask is the mask used on the remote network, not the source network. Since there is no number at the end of the static route, it is using the default administrative distance of 1
